Part 1:

Line	Instruction	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
3	CMP A, D	IF	ID	EX	MEM	WB																
4	BRGE		IF	ID	**	EX	MEM	WB														
5	LOAD D,			IF	**	ID	EX	MEM	WB													
6	SUB D, A				IF	**	ID	EX	MEM	WB												
7	CMP B, D					IF	**	ID	EX	MEM	WB											
8	BRGE						IF	**	ID	**	EX	MEM	WB									
9	LOADF C, [array+B]							IF	**	**	ID	EX	MEM	WB								
10.	LOADF D, [array+B+1]								IF	**	**	ID	EX	MEM	WB							
11.	CMP D, C									IF	**	**	ID	EX	MEM	WB						
12.	BRGE										IF	**	**	ID	**	EX	MEM	WB				
13	STOREF [array+B], D											IF	**	**	**	ID	EX	MEM	WB			
14.	STOREF [array+B+1], C												IF	**	**	**	ID	EX	MEM	WB		
15	ADDI B, 1													IF	**	**	**	ID	EX	MEM	WB	
16.	JUMP														IF	**	**	**	ID	EX	MEM	WB

If at line 12 (BRGE) taken – elements are already in order, so no swap:

		1	2	3	4	5	6	7	8	9
11.	CMP D, C	IF	ID	EX	MEM	WB				
12	BRGE		IF	ID	**	EX	MEM	WB		
13.	STOREF [array+B], D (flushed)			IF	**	XX				
15	ADDI B, 1				IF	**	ID	EX	MEM	WB

If at line 8 (BRGE) taken – inner loop finished.

		1	2	3	4	5	6	7	8	9
7	CMP B, D	IF	ID	EX	MEM	WB				
8	BRGE		IF	ID	**	EX	MEM	WB		
9	LOADF C, [array+B]			IF	**	XX				

If at line 4 (BRGE) taken – whole sort finished.

		1	2	3	4	5	6	7	8	9
3.	CMP A, D	IF	ID	EX	MEM	WB				
4.	BRGE		IF	ID	**	EX	MEM	WB		
5	LOAD D, [last]			IF	**	XX				

Part 2:

Hazard Table

Instruction pair	Hazard type	Cause	Resolution
CMP → BRGE (3 → 4)	Flags RAW	C14 writes flags, branch reads flags	1 stall + forward
LOAD → SUB (5 → 6)	Load-use RAW	SUB needs loaded D	MEM/WB → EX forwarding
SUB → CMP (6 → 7)	RAW	CMP needs D	EX forwarding
CMP → BRGE (7 → 8)	Flags RAW	Branch reads flags	1 stall + forward
LOADF C → CMP (9 → 11)	RAW	CMP needs C	Forward MEM/WB → EX
LOADF D → CMP (10 → 11)	RAW	CMP needs D	Forward MEM/WB → EX
CMP → BRGE (11 → 12)	Flags RAW	Branch uses flags	1 stall + forwarding
BRGE taken	Control	Wrong instructions fetched	Flush IF/ID
JUMP inner	Control	PC changes	Flush IF/ID